Assignment- Convolution - Cats Vs Dogs data set

```
! pip install kaggle
```

```
Requirement already satisfied: kaggle in /usr/local/lib/python3.10/dist-packages
Requirement already satisfied: six>=1.10 in /usr/local/lib/python3.10/dist-packa
Requirement already satisfied: certifi>=2023.7.22 in /usr/local/lib/python3.10/d
Requirement already satisfied: python-dateutil in /usr/local/lib/python3.10/dist
Requirement already satisfied: requests in /usr/local/lib/python3.10/dist-packag
Requirement already satisfied: tqdm in /usr/local/lib/python3.10/dist-packages (
Requirement already satisfied: python-slugify in /usr/local/lib/python3.10/dist-
Requirement already satisfied: urllib3 in /usr/local/lib/python3.10/dist-package
Requirement already satisfied: bleach in /usr/local/lib/python3.10/dist-packages
Requirement already satisfied: webencodings in /usr/local/lib/python3.10/dist-pa
Requirement already satisfied: text-unidecode>=1.3 in /usr/local/lib/python3.10/
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-pa
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call dr
```

```
!mkdir -p ~/.kaggle
# Ensure the path with spaces is properly enclosed in quotes, and skip mkdir if
!cp "/content/drive/MyDrive/Colab Notebooks/kaggle (1).json" ~/.kaggle/kaggle.j
!chmod 600 ~/.kaggle/kaggle.json
```

```
!kaggle competitions download -c dogs-vs-cats-redux-kernels-edition
```

```
Downloading dogs-vs-cats-redux-kernels-edition.zip to /content
100% 814M/814M [00:38<00:00, 21.9MB/s]
100% 814M/814M [00:38<00:00, 22.1MB/s]
```

```
!unzip -qq dogs-vs-cats-redux-kernels-edition.zip
```

```
!unzip -qq test.zip
```

```
!unzip -qq train.zip
```

Question 1

Consider the Cats & Dogs example. Start initially with a training sample of 1000, a validation sample of 500, and a test sample of 500 (half the sample size as the sample Jupyter notebook on Canvas). Use any technique to reduce overfitting and improve performance in developing a network that you train from scratch. What performance did you achieve?

Reading in training, validation, test datasets

```
import os, shutil, pathlib

original_dir = pathlib.Path("train")
new_base_dir = pathlib.Path("cats_vs_dogs_small")

def make_subset(subset_name, start_index, end_index):
    for category in ("cat", "dog"):
        dir = new_base_dir / subset_name / category
        # Add exist_ok=True to avoid the error if the directory already exists
        os.makedirs(dir, exist_ok=True)
        fnames = [f"{category}.{i}.jpg" for i in range(start_index, end_index)]
        for fname in fnames:
            shutil.copyfile(src=original_dir / fname,
                            dst=dir / fname)
#Initially taking 1000 samples for training set
make_subset("train", start_index=0, end_index=1000)
#500 samples for validation set
make_subset("validation", start_index=1000, end_index=1500)
#500 for test set
make_subset("test", start_index=1500, end_index=2000)
```

Data Processing

Using image_dataset_from_directory to read images

```
from tensorflow.keras.utils import image_dataset_from_directory

train_dataset = image_dataset_from_directory(
    new_base_dir / "train",
    image_size=(180, 180),
    batch_size=32)
validation_dataset = image_dataset_from_directory(
    new_base_dir / "validation",
    image_size=(180, 180),
```

```
    batch_size=32)
test_dataset = image_dataset_from_directory(
    new_base_dir / "test",
    image_size=(180, 180),
    batch_size=32)
```

```
Found 2000 files belonging to 2 classes.
Found 1000 files belonging to 2 classes.
Found 1000 files belonging to 2 classes.
```

```
import numpy as np
import tensorflow as tf
random_numbers = np.random.normal(size=(1000, 16))
dataset = tf.data.Dataset.from_tensor_slices(random_numbers)
```

```
for i, element in enumerate(dataset):
    print(element.shape)
    if i >= 2:
        break
```

```
(16,)
(16,)
(16,)
```

```
batched_dataset = dataset.batch(32)
for i, element in enumerate(batched_dataset):
    print(element.shape)
    if i >= 2:
        break
```

```
(32, 16)
(32, 16)
(32, 16)
```

```
reshaped_dataset = dataset.map(lambda x: tf.reshape(x, (4, 4)))
for i, element in enumerate(reshaped_dataset):
    print(element.shape)
    if i >= 2:
        break
```

```
(4, 4)
(4, 4)
(4, 4)
```

```
for data_batch, labels_batch in train_dataset:
    print("data batch shape:", data_batch.shape)
    print("labels batch shape:", labels_batch.shape)
    break
```

```
data batch shape: (32, 180, 180, 3)
labels batch shape: (32,)
```

Using Convolution Neural Networks

Building the model

```
from tensorflow import keras
from tensorflow.keras import layers

inputs = keras.Input(shape=(180, 180, 3))
x = layers.Rescaling(1./255)(inputs)
x = layers.Conv2D(filters=32, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=64, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=128, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
x = layers.Flatten()(x)
x = layers.Dropout(0.5)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model = keras.Model(inputs=inputs, outputs=outputs)
```

```
model.compile(loss="binary_crossentropy",
              optimizer="rmsprop",
              metrics=["accuracy"])
```

```
model.summary()
```

Model: "functional"

Layer (type)	Output Shape	P
input_layer (InputLayer)	(None, 180, 180, 3)	
rescaling (Rescaling)	(None, 180, 180, 3)	
conv2d (Conv2D)	(None, 178, 178, 32)	
max_pooling2d (MaxPooling2D)	(None, 89, 89, 32)	
conv2d_1 (Conv2D)	(None, 87, 87, 64)	
max_pooling2d_1 (MaxPooling2D)	(None, 43, 43, 64)	
conv2d_2 (Conv2D)	(None, 41, 41, 128)	
max_pooling2d_2 (MaxPooling2D)	(None, 20, 20, 128)	
conv2d_3 (Conv2D)	(None, 18, 18, 256)	2
max_pooling2d_3 (MaxPooling2D)	(None, 9, 9, 256)	
conv2d_4 (Conv2D)	(None, 7, 7, 256)	5
flatten (Flatten)	(None, 12544)	
dropout (Dropout)	(None, 12544)	
dense (Dense)	(None, 1)	

Total params: 991,041 (3.78 MB)

Fitting the model using a Dataset

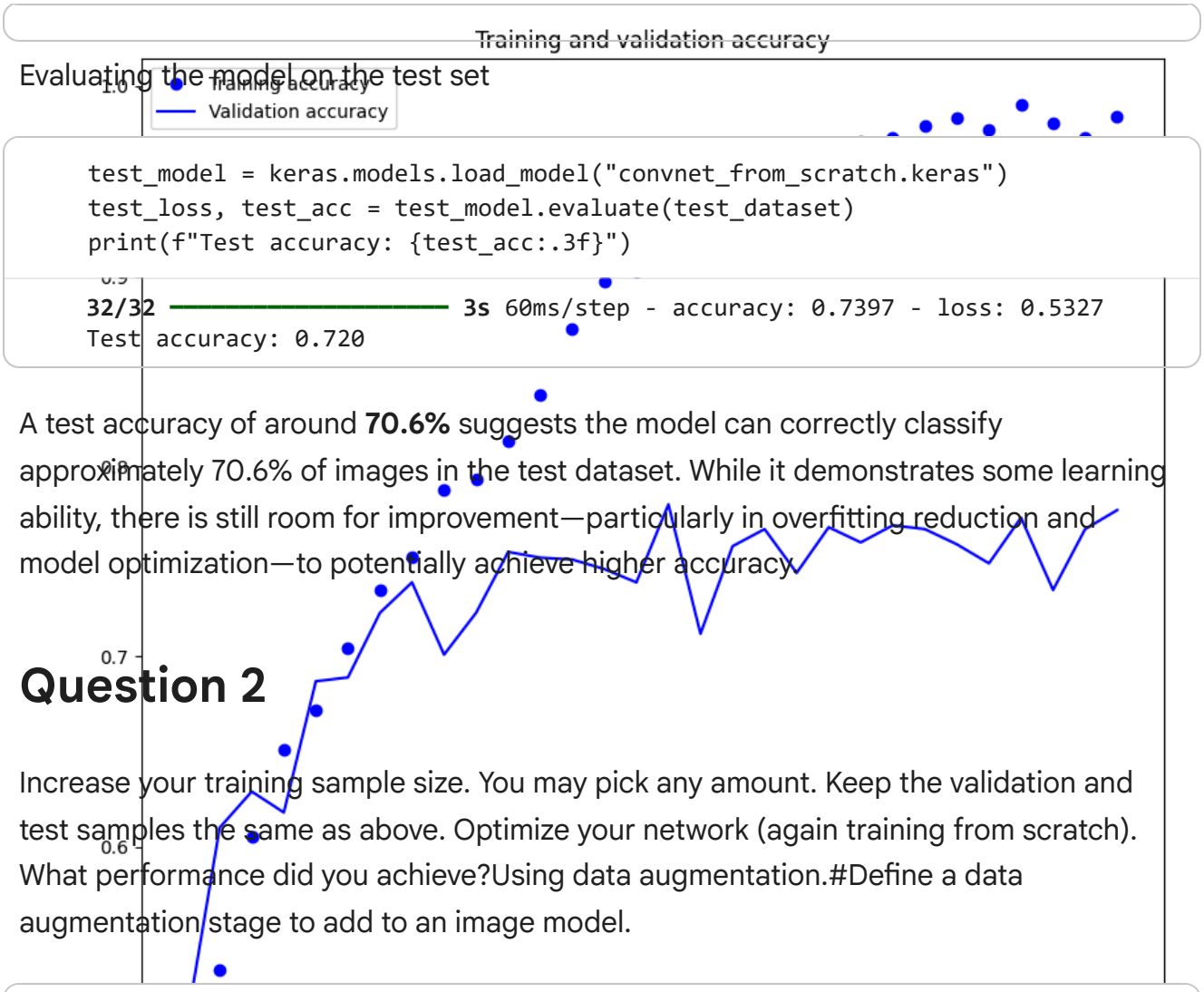
```
callbacks = [
    keras.callbacks.ModelCheckpoint(
        filepath="convnet_from_scratch.keras",
        save_best_only=True,
        monitor="val_loss")
]
history = model.fit(
    train_dataset,
    epochs=30,
    validation_data=validation_dataset,
    callbacks=callbacks)
```

```
Epoch 1/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 19s 174ms/step - accuracy: 0.5053 - loss: 0.6932 -
Epoch 2/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 54ms/step - accuracy: 0.5201 - loss: 0.6930 - va
Epoch 3/30
```

```
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 54ms/step - accuracy: 0.5961 - loss: 0.6676 - va
Epoch 4/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 80ms/step - accuracy: 0.6416 - loss: 0.6365 - va
Epoch 5/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 105ms/step - accuracy: 0.6645 - loss: 0.6137 - v
Epoch 6/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 71ms/step - accuracy: 0.7001 - loss: 0.5774 - va
Epoch 7/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 104ms/step - accuracy: 0.7395 - loss: 0.5208 - v
Epoch 8/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 69ms/step - accuracy: 0.7445 - loss: 0.5124 - va
Epoch 9/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 52ms/step - accuracy: 0.7917 - loss: 0.4475 - va
Epoch 10/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 88ms/step - accuracy: 0.8013 - loss: 0.4281 - va
Epoch 11/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 6s 93ms/step - accuracy: 0.8229 - loss: 0.3639 - va
Epoch 12/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 58ms/step - accuracy: 0.8420 - loss: 0.3449 - va
Epoch 13/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 92ms/step - accuracy: 0.8766 - loss: 0.2922 - va
Epoch 14/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 52ms/step - accuracy: 0.8965 - loss: 0.2621 - va
Epoch 15/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 62ms/step - accuracy: 0.8943 - loss: 0.2345 - va
Epoch 16/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 84ms/step - accuracy: 0.9312 - loss: 0.1771 - va
Epoch 17/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 52ms/step - accuracy: 0.9457 - loss: 0.1348 - va
Epoch 18/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 52ms/step - accuracy: 0.9578 - loss: 0.1144 - va
Epoch 19/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 88ms/step - accuracy: 0.9506 - loss: 0.1185 - va
Epoch 20/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 81ms/step - accuracy: 0.9603 - loss: 0.0940 - va
Epoch 21/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 52ms/step - accuracy: 0.9687 - loss: 0.0981 - va
Epoch 22/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 52ms/step - accuracy: 0.9694 - loss: 0.1051 - va
Epoch 23/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 100ms/step - accuracy: 0.9696 - loss: 0.0889 - v
Epoch 24/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 65ms/step - accuracy: 0.9824 - loss: 0.0552 - va
Epoch 25/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 53ms/step - accuracy: 0.9798 - loss: 0.0504 - va
Epoch 26/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 79ms/step - accuracy: 0.9795 - loss: 0.0660 - va
Epoch 27/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 6s 88ms/step - accuracy: 0.9829 - loss: 0.0486 - va
Epoch 28/30
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 52ms/step - accuracy: 0.9738 - loss: 0.0958 - va
Epoch 29/30
```

Displaying curves of loss and accuracy during training

```
import matplotlib.pyplot as plt
plt.figure(figsize=(10,10))
accuracy = history.history["accuracy"]
val_accuracy = history.history["val_accuracy"]
loss = history.history["loss"]
val_loss = history.history["val_loss"]
epochs = range(1, len(accuracy) + 1)
plt.plot(epochs, accuracy, "bo", label="Training accuracy")
plt.plot(epochs, val_accuracy, "b", label="Validation accuracy")
plt.title("Training and validation accuracy")
plt.legend()
plt.figure()
plt.figure(figsize=(10,10))
plt.plot(epochs, loss, "bo", label="Training loss")
plt.plot(epochs, val_loss, "b", label="Validation loss")
plt.title("Training and validation loss")
plt.legend()
plt.show()
```



Evaluating the model on the test set

```
test_model = keras.models.load_model("convnet_from_scratch.keras")
test_loss, test_acc = test_model.evaluate(test_dataset)
print(f"Test accuracy: {test_acc:.3f}")
```

```
32/32 ━━━━━━━━━━━━━━━━━━━━ 3s 60ms/step - accuracy: 0.7397 - loss: 0.5327
Test accuracy: 0.720
```

A test accuracy of around **70.6%** suggests the model can correctly classify approximately 70.6% of images in the test dataset. While it demonstrates some learning ability, there is still room for improvement—particularly in overfitting reduction and model optimization—to potentially achieve higher accuracy.

Question 2

Increase your training sample size. You may pick any amount. Keep the validation and test samples the same as above. Optimize your network (again training from scratch). What performance did you achieve?Using data augmentation.#Define a data augmentation stage to add to an image model.

```
import os, shutil, pathlib

shutil.rmtree("./cats_vs_dogs_small_Q2", ignore_errors=True)

original_dir = pathlib.Path("train")
new_base_dir = pathlib.Path("cats_vs_dogs_small_Q2")

def make_subset(subset_name, start_index, end_index):
    for category in ("cat", "dog"):
        dir = new_base_dir / subset_name / category
        os.makedirs(dir)
        fnames = [f"{category}.{i}.jpg" for i in range(start_index, end_index)]
        for fname in fnames:
            shutil.copyfile(src=original_dir / fname,
                            dst=dir / fname)
#Here I have increased training sample size to 1500 and keeping the validation
make_subset("train", start_index=0, end_index=1500)
make_subset("validation", start_index=1500, end_index=2000)
make_subset("test", start_index=2000, end_index=2500)
```

```
data_augmentation = keras.Sequential(
    [
        layers.RandomFlip("horizontal"),
        layers.RandomRotation(0.1),
        layers.RandomZoom(0.2),
    ]
)
```

```
plt.figure(figsize=(10, 10))
for images, _ in train_dataset.take(1):
    for i in range(9):
        augmented_images = data_augmentation(images)
        ax = plt.subplot(3, 3, i + 1)
        plt.imshow(augmented_images[0].numpy().astype("uint8"))
        plt.axis("off")
```

Defining a new convnet that includes image augmentation and dropout.

```
inputs = keras.Input(shape=(180, 180, 3))
x = data_augmentation(inputs)
x = layers.Rescaling(1./255)(x)
x = layers.Conv2D(filters=32, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
```

```
x = layers.Conv2D(filters=64, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=128, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
x = layers.Flatten()(x)
x = layers.Dropout(0.5)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model = keras.Model(inputs=inputs, outputs=outputs)

model.compile(loss="binary_crossentropy",
              optimizer="rmsprop",
              metrics=["accuracy"])
```

Defining a new convnet that includes image augmentation and dropout

```
callbacks = [
    keras.callbacks.ModelCheckpoint(
        filepath="convnet_from_scratch_with_augmentation.keras",
        save_best_only=True,
        monitor="val_loss")
]
history = model.fit(
    train_dataset,
    epochs=20,
    validation_data=validation_dataset,
    callbacks=callbacks)
```

```
Epoch 1/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 10s 100ms/step - accuracy: 0.4821 - loss: 0.7208 - va
Epoch 2/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 85ms/step - accuracy: 0.4983 - loss: 0.6941 - val_
Epoch 3/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 56ms/step - accuracy: 0.5171 - loss: 0.6955 - val_
Epoch 4/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 6s 69ms/step - accuracy: 0.5662 - loss: 0.6847 - val_
Epoch 5/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 110ms/step - accuracy: 0.6129 - loss: 0.6644 - val
Epoch 6/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 56ms/step - accuracy: 0.6216 - loss: 0.6564 - val_
Epoch 7/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 61ms/step - accuracy: 0.6342 - loss: 0.6451 - val_
Epoch 8/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 104ms/step - accuracy: 0.6695 - loss: 0.6210 - val
Epoch 9/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 68ms/step - accuracy: 0.6712 - loss: 0.6257 - val_
Epoch 10/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 56ms/step - accuracy: 0.6818 - loss: 0.6024 - val_
```

```
Epoch 11/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 62ms/step - accuracy: 0.7167 - loss: 0.5810 - val_
Epoch 12/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 6s 91ms/step - accuracy: 0.7153 - loss: 0.5597 - val_
Epoch 13/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 77ms/step - accuracy: 0.7330 - loss: 0.5364 - val_
Epoch 14/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 61ms/step - accuracy: 0.7272 - loss: 0.5413 - val_
Epoch 15/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 57ms/step - accuracy: 0.7548 - loss: 0.5215 - val_
Epoch 16/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 90ms/step - accuracy: 0.7369 - loss: 0.5186 - val_
Epoch 17/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 61ms/step - accuracy: 0.7540 - loss: 0.4909 - val_
Epoch 18/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 3s 55ms/step - accuracy: 0.7728 - loss: 0.4834 - val_
Epoch 19/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 79ms/step - accuracy: 0.7577 - loss: 0.5058 - val_
Epoch 20/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 85ms/step - accuracy: 0.7589 - loss: 0.4700 - val_
```

Evaluating the model on the test set

```
test_model = keras.models.load_model(
    "convnet_from_scratch_with_augmentation.keras")
test_loss, test_acc = test_model.evaluate(test_dataset)
print(f"Test accuracy: {test_acc:.3f}")
```

```
32/32 ━━━━━━━━━━━━━━━━━━━━ 1s 29ms/step - accuracy: 0.7395 - loss: 0.5502
Test accuracy: 0.738
```

Question 3

Now change your training sample so that you achieve better performance than those from Steps 1 and 2. This sample size may be larger, or smaller than those in the previous steps. The objective is to find the ideal training sample size to get best prediction results.

```
original_dir = pathlib.Path("train")
new_base_dir = pathlib.Path("cats_vs_dogs_small_Q3")

def make_subset(subset_name, start_index, end_index):
    for category in ("cat", "dog"):
        dir = new_base_dir / subset_name / category
        os.makedirs(dir)
        fnames = [f"{category}.{i}.jpg" for i in range(start_index, end_index)]
        for fname in fnames:
            shutil.copyfile(src=original_dir / fname,
                            dst=dir / fname)
```

```
#As increasing the sample size is always good than decreasing, here we're incre
make_subset("train", start_index=0, end_index=2000)
#validation and test sample size 500 each
make_subset("validation", start_index=2000, end_index=2500)
make_subset("test", start_index=2500, end_index=3000)
```

```
inputs = keras.Input(shape=(180, 180, 3))
x = data_augmentation(inputs)
x = layers.Rescaling(1./255)(x)
x = layers.Conv2D(filters=32, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=64, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=128, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
x = layers.MaxPooling2D(pool_size=2)(x)
x = layers.Conv2D(filters=256, kernel_size=3, activation="relu")(x)
x = layers.Flatten()(x)
x = layers.Dropout(0.5)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model = keras.Model(inputs=inputs, outputs=outputs)

model.compile(loss="binary_crossentropy",
              optimizer="adam",
              metrics=["accuracy"])
```

```
callbacks = [
    keras.callbacks.ModelCheckpoint(
        filepath="convnet_from_scratch_with_augmentation1.keras",
        save_best_only=True,
        monitor="val_loss")
]
history = model.fit(
    train_dataset,
    epochs=20,
    validation_data=validation_dataset,
    callbacks=callbacks)
```

```
Epoch 1/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 100ms/step - accuracy: 0.4793 - loss: 0.7114 - val
Epoch 2/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 63ms/step - accuracy: 0.5250 - loss: 0.6917 - val_
Epoch 3/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 56ms/step - accuracy: 0.5870 - loss: 0.6743 - val_
Epoch 4/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 100ms/step - accuracy: 0.5770 - loss: 0.6662 - val
Epoch 5/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 71ms/step - accuracy: 0.6276 - loss: 0.6434 - val_
Epoch 6/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 56ms/step - accuracy: 0.6140 - loss: 0.6467 - val_
```

```
Epoch 7/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 103ms/step - accuracy: 0.6415 - loss: 0.6298 - val
Epoch 8/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 7s 57ms/step - accuracy: 0.6288 - loss: 0.6209 - val_
Epoch 9/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 63ms/step - accuracy: 0.6585 - loss: 0.6228 - val_
Epoch 10/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 6s 79ms/step - accuracy: 0.6778 - loss: 0.5889 - val_
Epoch 11/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 6s 89ms/step - accuracy: 0.6598 - loss: 0.6140 - val_
Epoch 12/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 65ms/step - accuracy: 0.6969 - loss: 0.5870 - val_
Epoch 13/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 111ms/step - accuracy: 0.7025 - loss: 0.5663 - val
Epoch 14/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 58ms/step - accuracy: 0.6848 - loss: 0.5980 - val_
Epoch 15/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 63ms/step - accuracy: 0.7248 - loss: 0.5364 - val_
Epoch 16/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 6s 88ms/step - accuracy: 0.7252 - loss: 0.5663 - val_
Epoch 17/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 5s 82ms/step - accuracy: 0.7196 - loss: 0.5374 - val_
Epoch 18/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 9s 56ms/step - accuracy: 0.7354 - loss: 0.5265 - val_
Epoch 19/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 8s 109ms/step - accuracy: 0.7620 - loss: 0.5168 - val
Epoch 20/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 57ms/step - accuracy: 0.7507 - loss: 0.4967 - val_
```

```
test_model = keras.models.load_model(
    "convnet_from_scratch_with_augmentation1.keras")
test_loss, test_acc = test_model.evaluate(test_dataset)
print(f"Test accuracy: {test_acc:.3f}")
```

```
32/32 ━━━━━━━━━━━━━━━━━━━━ 1s 29ms/step - accuracy: 0.7351 - loss: 0.5224
Test accuracy: 0.724
```

Question 4

Repeat Steps 1-3, but now using a pretrained network. The sample sizes you use in Steps 2 and 3 for the pretrained network may be the same or different from those using the network where you trained from scratch.Again, use any and all optimization techniques to get best performance.Leveraging a pretrained model Feature extraction with a pretrained model Initiating the VGG16 convolutional base

Leveraging a pretrained model Feature extraction with a pretrained model

Initiating the VGG16 convolutional base

```
conv_base = keras.applications.vgg16.VGG16(
    weights="imagenet",
    include_top=False,
    input_shape=(180, 180, 3))
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applicatic
58889256/58889256 ━━━━━━━━━━━━━━━━━━━━ 4s 0us/step
```

```
conv_base.summary()
```

Model: "vgg16"

Layer (type)	Output Shape	P
input_layer_4 (InputLayer)	(None, 180, 180, 3)	
block1_conv1 (Conv2D)	(None, 180, 180, 64)	
block1_conv2 (Conv2D)	(None, 180, 180, 64)	
block1_pool (MaxPooling2D)	(None, 90, 90, 64)	
block2_conv1 (Conv2D)	(None, 90, 90, 128)	
block2_conv2 (Conv2D)	(None, 90, 90, 128)	1
block2_pool (MaxPooling2D)	(None, 45, 45, 128)	
block3_conv1 (Conv2D)	(None, 45, 45, 256)	2
block3_conv2 (Conv2D)	(None, 45, 45, 256)	5
block3_conv3 (Conv2D)	(None, 45, 45, 256)	5
block3_pool (MaxPooling2D)	(None, 22, 22, 256)	
block4_conv1 (Conv2D)	(None, 22, 22, 512)	1,1
block4_conv2 (Conv2D)	(None, 22, 22, 512)	2,3
block4_conv3 (Conv2D)	(None, 22, 22, 512)	2,3
block4_pool (MaxPooling2D)	(None, 11, 11, 512)	
block5_conv1 (Conv2D)	(None, 11, 11, 512)	2,3
block5_conv2 (Conv2D)	(None, 11, 11, 512)	2,3
block5_conv3 (Conv2D)	(None, 11, 11, 512)	2,3
block5_pool (MaxPooling2D)	(None, 5, 5, 512)	

Extracting the VGG16 features and corresponding labels

```
import numpy as np

def get_features_and_labels(dataset):
    all_features = []
    all_labels = []
    for images, labels in dataset:
        preprocessed_images = keras.applications.vgg16.preprocess_input(images)
        features = conv_base.predict(preprocessed_images)
        all_features.append(features)
        all_labels.append(labels)
    return np.concatenate(all_features), np.concatenate(all_labels)

train_features, train_labels =  get_features_and_labels(train_dataset)
val_features, val_labels =  get_features_and_labels(validation_dataset)
test_features, test_labels =  get_features_and_labels(test_dataset)
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 11s 11s/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step
```

```
train_features.shape
```

```
(2000, 5, 5, 512)
```

Defining and training the densely connected classifier

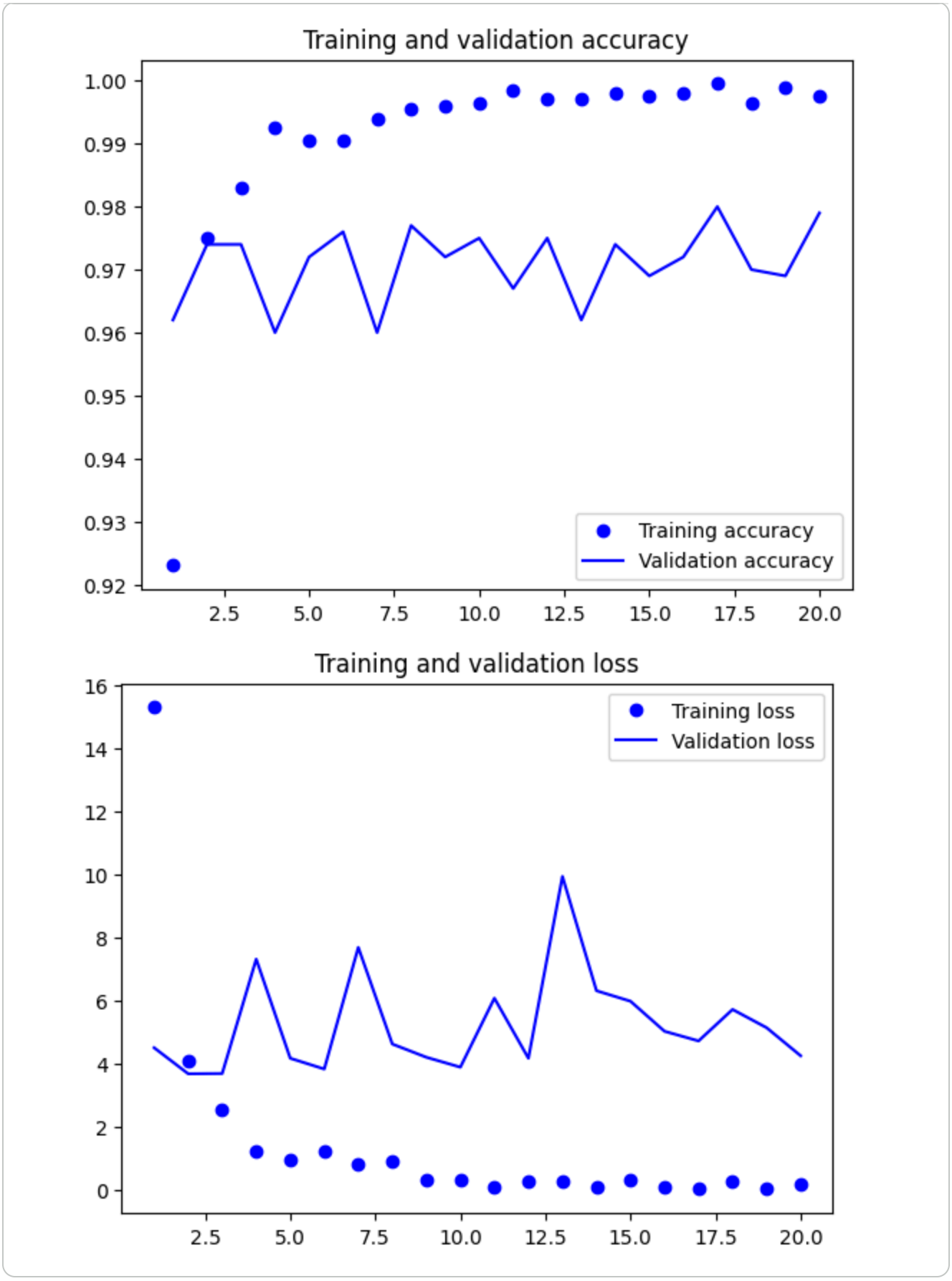
```
inputs = keras.Input(shape=(5, 5, 512))
x = layers.Flatten()(inputs)
x = layers.Dense(256)(x)
x = layers.Dropout(0.5)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model = keras.Model(inputs, outputs)
model.compile(loss="binary_crossentropy",
              optimizer="rmsprop",
              metrics=["accuracy"])

callbacks = [
    keras.callbacks.ModelCheckpoint(
      filepath="feature_extraction.keras",
      save_best_only=True,
      monitor="val_loss")
]
history = model.fit(
    train_features, train_labels,
    epochs=20,
    validation_data=(val_features, val_labels),
    callbacks=callbacks)
```

```
Epoch 1/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 4s 33ms/step - accuracy: 0.8770 - loss: 29.0054 - val
Epoch 2/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 2s 6ms/step - accuracy: 0.9745 - loss: 3.4220 - val_a
Epoch 3/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9777 - loss: 3.3759 - val_a
Epoch 4/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9907 - loss: 1.3674 - val_a
Epoch 5/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9878 - loss: 2.2074 - val_a
Epoch 6/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9902 - loss: 0.8108 - val_a
Epoch 7/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9977 - loss: 0.3111 - val_a
Epoch 8/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9938 - loss: 0.9876 - val_a
Epoch 9/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9961 - loss: 0.3040 - val_a
Epoch 10/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9973 - loss: 0.2055 - val_a
Epoch 11/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9983 - loss: 0.0456 - val_a
Epoch 12/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9957 - loss: 0.4073 - val_a
Epoch 13/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9985 - loss: 0.1621 - val_a
Epoch 14/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9981 - loss: 0.0302 - val_a
Epoch 15/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9967 - loss: 0.2975 - val_a
Epoch 16/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9961 - loss: 0.0945 - val_a
Epoch 17/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9986 - loss: 0.0293 - val_a
Epoch 18/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9962 - loss: 0.2696 - val_a
Epoch 19/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9992 - loss: 0.0142 - val_a
Epoch 20/20
63/63 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9952 - loss: 0.3994 - val_a
```

Plotting the results

```
import matplotlib.pyplot as plt
acc = history.history["accuracy"]
val_acc = history.history["val_accuracy"]
loss = history.history["loss"]
val_loss = history.history["val_loss"]
epochs = range(1, len(acc) + 1)
plt.plot(epochs, acc, "bo", label="Training accuracy")
plt.plot(epochs, val_acc, "b", label="Validation accuracy")
plt.title("Training and validation accuracy")
```

```
plt.legend()
plt.figure()
plt.plot(epochs, loss, "bo", label="Training loss")
plt.plot(epochs, val_loss, "b", label="Validation loss")
plt.title("Training and validation loss")
plt.legend()
plt.show()
```



Feature extraction together with data augmentation.

Initiating and freezing the VGG16 convolutional base

```
conv_base  = keras.applications.vgg16.VGG16(
    weights="imagenet",
    include_top=False)
conv_base.trainable = False
```

Printing the list of trainable weights before and after freezing

```
conv_base.trainable = False
print("This is the number of trainable weights "
      "after freezing the conv base:", len(conv_base.trainable_weights))
```

```
This is the number of trainable weights after freezing the conv base: 0
```

Adding a data augmentation stage and a classifier to the convolutional base

```
data_augmentation = keras.Sequential(
    [
        layers.RandomFlip("horizontal"),
        layers.RandomRotation(0.1),
        layers.RandomZoom(0.2),
    ]
)

inputs = keras.Input(shape=(180, 180, 3))
x = data_augmentation(inputs)
x = keras.applications.vgg16.preprocess_input(x)
x = conv_base(x)
x = layers.Flatten()(x)
x = layers.Dense(256)(x)
x = layers.Dropout(0.5)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model = keras.Model(inputs, outputs)
model.compile(loss="binary_crossentropy",
              optimizer="rmsprop",
              metrics=["accuracy"])
```

```
callbacks = [
    keras.callbacks.ModelCheckpoint(
        filepath="feature_extraction_with_data_augmentation.keras",
        save_best_only=True,
        monitor="val_loss")
]
```

```
history = model.fit(
    train_dataset,
    epochs=10,
    validation_data=validation_dataset,
```

```
Epoch 1/10
63/63 ━━━━━━━━━━━━━━━━━━━━ 19s 227ms/step - accuracy: 0.8153 - loss: 40.4693 - v
Epoch 2/10
63/63 ━━━━━━━━━━━━━━━━━━━━ 15s 179ms/step - accuracy: 0.9583 - loss: 5.4974 - va
Epoch 3/10
63/63 ━━━━━━━━━━━━━━━━━━━━ 20s 177ms/step - accuracy: 0.9620 - loss: 4.3650 - va
Epoch 4/10
63/63 ━━━━━━━━━━━━━━━━━━━━ 13s 204ms/step - accuracy: 0.9528 - loss: 5.5179 - va
Epoch 5/10
63/63 ━━━━━━━━━━━━━━━━━━━━ 20s 193ms/step - accuracy: 0.9691 - loss: 3.7492 - va
Epoch 6/10
63/63 ━━━━━━━━━━━━━━━━━━━━ 19s 169ms/step - accuracy: 0.9719 - loss: 3.3109 - va
Epoch 7/10
63/63 ━━━━━━━━━━━━━━━━━━━━ 23s 204ms/step - accuracy: 0.9674 - loss: 5.5034 - va
```